

Single Board Computers for IoT

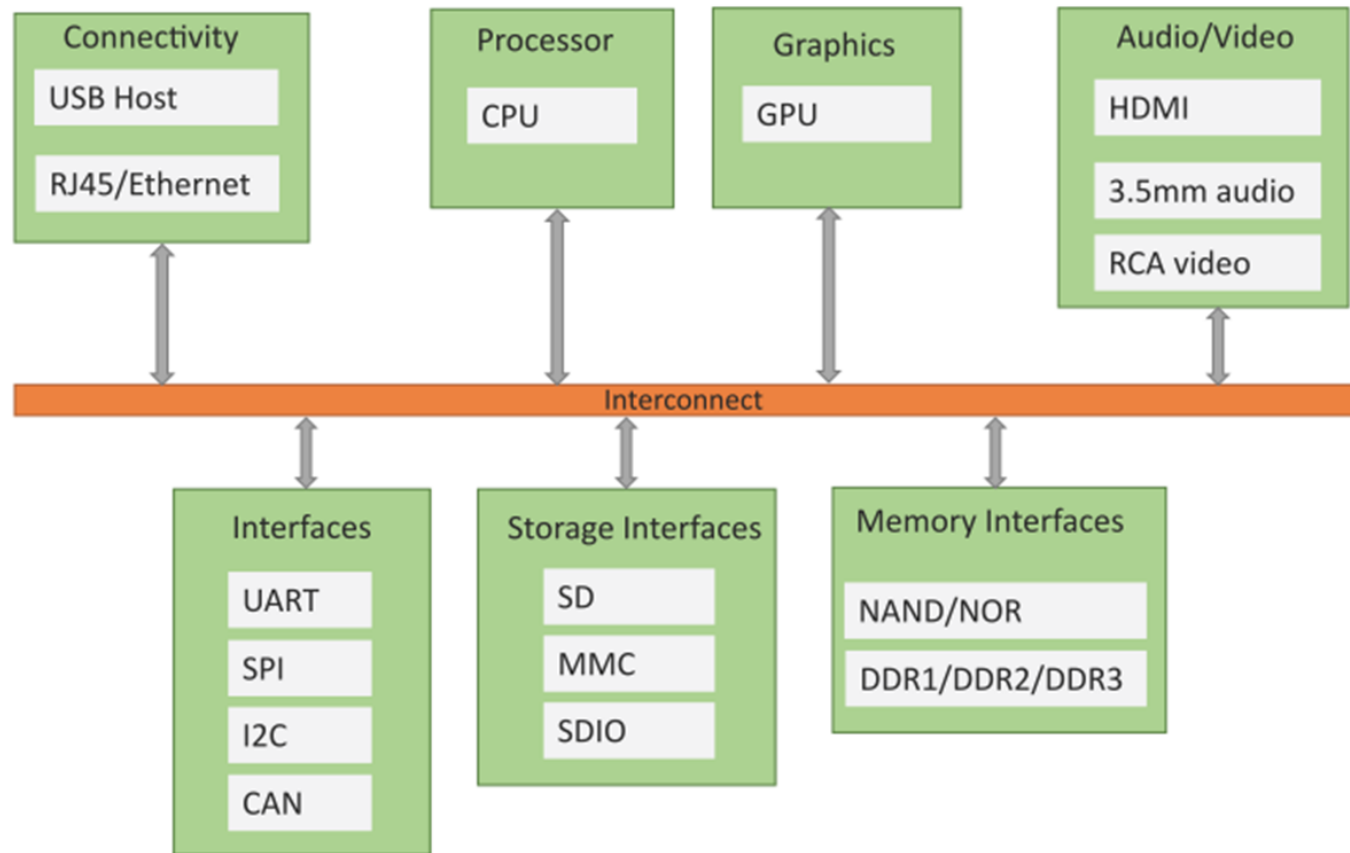
Agenda

- What is a Single Board Computer?
- Various type of SBCs

Single Board Computers (SBC)

- A complete, functioning computer with all functions (I/O, processor, memory) located on one board.
- They are lighter in weight, compact in size, more reliable and much more power efficient than multi-board computers.
- Single-board computers are mostly used in embedded applications.
- They are typically used in communication products or for a very specific applications that benefit from a pre-verified hardware and software combination to implement an industry standard.

Block Diagram of an IoT Device

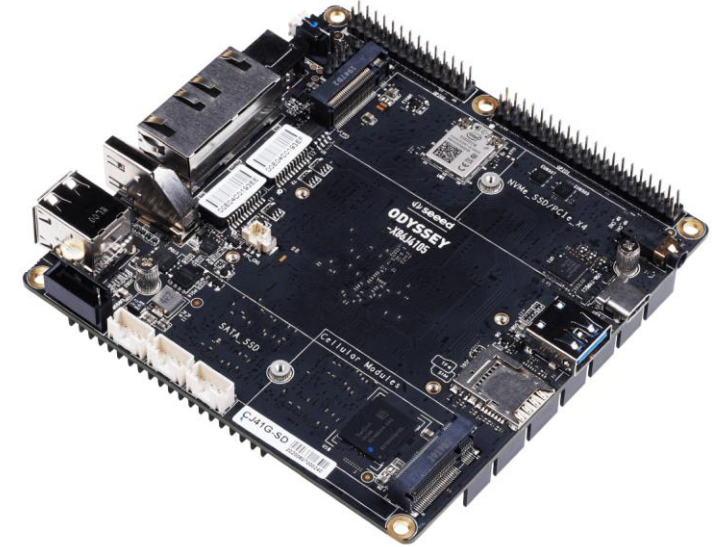


ODYSSEY - X86J

A powerful x86-based SBC designed for IoT, edge computing, and industrial applications.

Supports Windows, Linux, and various expansion modules for versatile development.

- Features:
 - Intel x86 CPU, up to 8GB RAM
 - Multiple USB, HDMI, and Ethernet ports.
 - Supports Wi-Fi, Bluetooth, and PCIe expansion
- Different Models/New Versions:
ODYSSEY - X86J4125, X86J4105 with different RAM/storage options



NVIDIA Jetson Nano Developer Kit

- A compact AI computing platform for deep learning, computer vision, and robotics.
- Supports CUDA, TensorRT, and AI frameworks for accelerated processing.
- Features:
 - Quad-core Cortex-A57 CPU, 128-core Maxwell
 - GPU4GB
 - LPDDR4 RAM,
 - microSD storage
 - Gigabit Ethernet, USB 3.0, and GPIO expansion
- Different Models/New Versions:
 - Jetson Xavier NX (more powerful alternative)



BeagleBone

A compact, open-source SBC with built-in Wi-Fi and Bluetooth for IoT projects.

- Features:
 - AM335x 1GHz ARM Cortex-A8 processor 512MB
 - DDR3 RAM,
 - Support Wi-Fi 802.11 b/g/n, Bluetooth 4.1
- Different Models/New Versions:
 - BeagleBone Black (wired version)
 - BeagleBone AI (enhanced AI capabilities)



Other players

- **Asus TinkerBoard**

- A high-performance SBC designed for makers and multimedia applications.

- **Banana Pi**

- A versatile SBC series offering Raspberry Pi alternatives with enhanced performance.

- **Orange Pi**

- A budget-friendly SBC series designed for IoT, AI, and embedded computing.

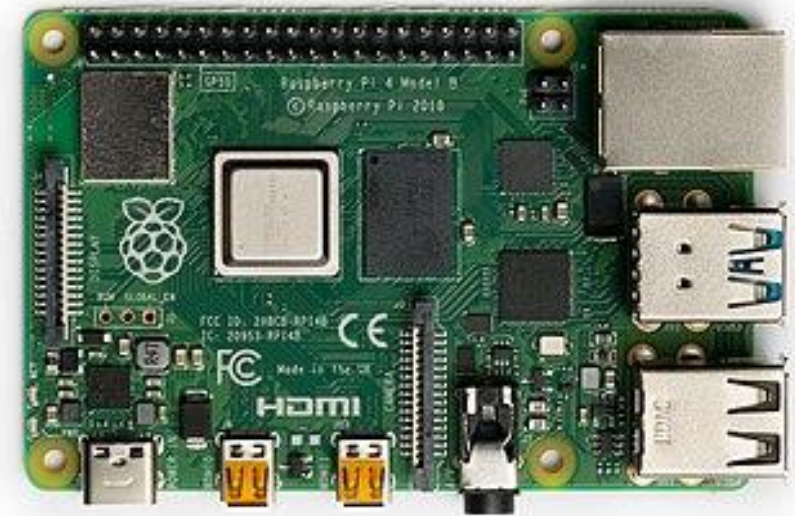
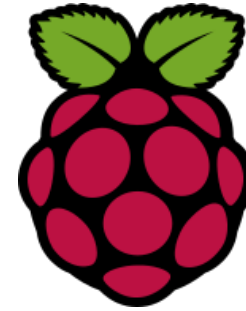
- **Odroid**

- A powerful SBC series with high-performance computing capabilities.



Raspberry Pi

- Raspberry Pi (RPi) is a low-cost mini-computer with the physical size of a credit card.
- Raspberry Pi runs various operating systems and can perform almost all the tasks that a normal desktop computer can do.
- Raspberry Pi also allows interfacing sensors and actuators through the general purpose I/O pins.



Raspberry Pi categories

RPis boards are categorized based on two items:

- Edition (Zero to 4) points to hardware generation and architecture
- Model (A, B, and Plus) points to the number of peripherals

- Raspberry Pi Zero and w
- Raspberry Pi
- Raspberry Pi 2
- Raspberry Pi 3
- Raspberry Pi 4/400
- Raspberry Pi Pico
- Raspberry Pi 5



RPi comparison

Specification	RPi 1 Model B	RPi 2 Model B	RPi 3 Model B	RPi 3 Model B+	RPi 4 Model B	RPi 400	RPi Zero W	RPi Zero 2 W	RPi 5
Release Year	2012	2015	2016	2018	2019	2020	2017	2021	2023
CPU	BCM2835, 700 MHz ARM11	BCM2836, 900 MHz Quad-A7	BCM2837, 1.2 GHz Quad-A53	BCM2837B0, 1.4 GHz Quad-A53	BCM2711, 1.5 GHz Quad-A72	BCM2711, 1.8 GHz Quad-A72	BCM2835, 1 GHz ARM11	BCM2710A1, 1 GHz Quad-A53	BCM2712, 2.4 GHz Quad-A76
RAM	256MB	1GB	1GB	1GB	2GB, 4GB, 8GB	4GB	512MB	512MB	4GB, 8GB
GPIO Pins	26	40	40	40	40	40 (via adapter)	40	40	40
Wi-Fi	No	No	Yes (802.11n)	Yes (802.11ac)	Yes (802.11ac)	Yes (802.11ac)	Yes (802.11n)	Yes (802.11n)	Yes (802.11ac)
Bluetooth	No	No	Yes (4.1)	Yes (4.2)	Yes (5.0)	Yes (5.0)	Yes (4.1)	Yes (4.2)	Yes (5.0)